Arista Access Layer QoS Congestion Mitigation (HEB-ACC-25)

Node: HEB-ACC-25 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9278

Description

This Standard Operating Procedure (SOP) addresses the mitigation of QoS congestion events, specifically microbursts, interface tail drops, and QoS queue starvation, on the Arista Access Layer device HEB-ACC-25. The underlying cause of these events stems from the interaction of various protocols and hardware components. In this context, the Arista device's Buffer-to-Interface (B2I) ratio, combined with the egress queuing mechanism, may lead to the occurrence of tail drops when the interface is overwhelmed with traffic. Furthermore, the QoS policy configuration, which governs the allocation of buffer space and bandwidth, may not be optimized for the specific traffic patterns observed on this link. As a result, the QoS queues may become starved, leading to packet drops and congestion. To mitigate these issues, it is essential to carefully analyze the traffic patterns, adjust the QoS policy configuration, and optimize the device's resource allocation to ensure that the interface and buffers are not overwhelmed.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible via remote hands or in-person presence.
- Confirm that the redundant link is operational and can assume traffic responsibility in case of failure.
- Ensure that the necessary Arista CLI credentials are available for login.

Potential Root Causes

Cause	Probability
Inadequate B2I ratio configuration	High
Insufficient QoS policy configuration	Medium
Overaggressive egress queuing strategy	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware buffer statistics, Expected Output: Buffer utilization, buffer-to-interface ratio, and buffer allocation summary, Description: This command isolates the buffer utilization and allocation summary, helping to identify potential issues with the B2I ratio and buffer space allocation.

Command: show qos policy summary, Expected Output: QoS policy configuration, buffer space allocation, and bandwidth allocation summary, Description: This command provides an overview of the QoS policy configuration, enabling the identification of potential issues with buffer space and bandwidth allocation.
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Command: show interface statistics, Expected Output: Interface statistics, including packet drops, errors, and throughput summary, Description: This command isolates the interface statistics, helping to identify potential issues with tail drops, packet errors, and throughput anomalies.
```

Resolution Steps

Step 1: Step: Adjust B2I Ratio Configuration, Command: config hardware buffer statistics b2i-ratio , Description: Adjust the B2I ratio to optimize buffer space allocation and prevent tail drops.

Step 2: Step: Optimize QoS Policy Configuration, Command: config qos policy buffer-space-allocation , Description: Adjust the QoS policy configuration to optimize buffer space allocation and prevent QoS queue starvation.

Step 3: Step: Implement Egress Queuing Strategy, Command: config qos policy egress-queuing-strategy , Description: Implement an egress queuing strategy that balances buffer allocation and packet throughput to prevent overaggressive queuing.

Post-Check Verification

```
Command: show hardware buffer statistics, Expected Output: Buffer utilization, buffer-to-interface ratio, and buffer allocation summary (optimized), Description: Verify that the buffer utilization and allocation summary are within acceptable limits after adjusting the B2I ratio configuration.

Command: show qos policy summary, Expected Output: QoS policy configuration, buffer space allocation, and bandwidth allocation summary (optimized), Description: Verify that the QoS policy configuration, buffer space allocation, and bandwidth allocation summary are within acceptable limits after optimizing the QoS policy configuration.

Command: show interface statistics, Expected Output: Interface statistics, including packet drops, errors, and throughput summary (optimized), Description: Verify that the interface statistics, including packet drops, errors, and throughput summary, are within acceptable limits after implementing the egress queuing strategy.
```

Roll-back Procedures

- Step: Revert B2I Ratio Configuration, Command: config hardware buffer statistics b2i-ratio , Description: Revert the B2I ratio configuration to its original value in case of failure.

- Step: Revert QoS Policy Configuration, Command: config qos policy buffer-space-allocation , Description: Revert the QoS policy configuration to its original value in case of failure.

- Step: Revert Egress Queuing Strategy, Command: config qos policy egress-queuing-strategy , Description: Revert the egress queuing strategy to its original value in case of failure.